MTP F26 · TEAM C · RADIO TEAM

Radio brief: band, module and link budget

Team Santiago, Andrian, Marcos **Version** 0.11 · Sun 18 Oct 2026 **Next session** Fri 23 Oct (week 5)

New in v0.11 (18 Oct): the micro is now a Raspberry Pi 3 (the team's request; its WiFi/BT chip disabled or desoldered, Atenea Q22). **Nothing changes in the radio**: same SP4, same omni, same margins, same NM spec. New for us: T0 also checks the Pi's own noise in our band and its current draw, and T2 measures the gap between frames on Linux (0.5 ms would give ~97 kbps instead of 107). Details on the Team C Pi 3 Switch page.

New in v0.10 (17 Oct): review fixes and the NM spec.

- The NM interop spec for all teams: radio registers, frame, gossip rules and a test vector.
- The NM channel moves to **RF_CH 77** (2477 MHz). Channel 80 was BLE advertising channel 39, and the hop set now skips all three BLE advertising channels: **37 channels, 30 ms hops**.
- A **conservative planning case**, worth reading before trusting the margins: MRM at 20 dBm keeps only +0.9 dB, and NM drops to −3.1 dB.
- Goodput is now ~107 kbps (the old model counted the header as data).
- The MRM near-far numbers include a +6 dB two-ray peak.

New in v0.9 (16 Oct): 20 dBm rechecked; two versions kept. Hopping (option A) stays the default because it meets EN 300 328 as written. **Option B** sends the same 20 dBm on one fixed channel without hopping: same margins, simpler firmware, but our narrow signal is 5–10 dB above the 10 mW/MHz density limit for non-hopping radios, so we use it only if the course explicitly says yes (Atenea Q11). Both are built and tested. See 20 dBm. Also new: an NM link-protocol diagram.

New in v0.8 (16 Oct): every team uses 20 dBm, so every team hops. Atenea Q3 confirmed 20 dBm in writing, and the other teams will use it too. For a narrowband radio that means hopping, so the fixed MRM channel plan is gone: our hop set now covers all 40 even channels (2402–2480 MHz), the teams share the band by listen-before-talk,

and the boxes need **≥ 10 m** between teams. Random hop collisions cost about a third of our ACKs in the worst case; the simulator still delivers the whole file in 81 s.

New in v0.7 (16 Oct): 20 dBm e.i.r.p. is the baseline for SRI and MRM. A professor said yes to 20 dBm, and Atenea Q3 confirmed it in writing. For our narrowband radio that is allowed only as a **frequency-hopping radio with listen-before-talk** (CNAF UN-85 a), EN 300 328 adaptive FHSS), so SRI and MRM now hop with the same SP4 and omni: **+10.5 dB on every link**, MRM 260 m **+13.9 dB**. NM stays at 10 mW on channel 80; Quick Mode uses the 10 mW channel too. See 20 dBm with frequency hopping.

New in v0.5 (decided 13 Oct, Santiago with Marcos): one omni antenna per box on the Ebyte E01-ML01SP4. Antennas cannot change between the competition modes, and the patches' risks in NM (aiming, the rim corner, tuning) outweighed their +1 dB on MRM. The SP4 has the antenna socket on the module, so the flex omni plugs straight in. Shopping list updated below.

New in v0.4: all four teams agreed to run **every mode at 2.4 GHz**, so 868 MHz is dropped. The radio is now the **Ebyte E01-ML01DP5** (Nordic nRF24L01+ with PA and LNA) at 250 kbps, with **two patch antennas** per box and our own channel in MRM. **Do not order the Adafruit RFM69 or the Molex antennas**; the new shopping list is below. **Order by Fri 16 Oct.**

Every figure carries a tag: `RULE` from the competition rules PDF, `VERIFIED` against a primary source, `ASSUMPTION` to be replaced by a measurement, and `ESTIMATE`.

Bottom line

1. **2.4 GHz is the right band for this competition.** In MRM all teams transmit at once `RULE`. At 868 MHz four teams would share one 250 kHz channel (about a quarter of 67 kbps each). At 2.4 GHz the four teams share 83.5 MHz by hopping, there is no duty-cycle limit, and NM has one common radio for everybody.
2. **Margin, now with 20 dBm.** SRI and MRM hop at up to 20 dBm e.i.r.p. (CNAF UN-85 a), approved by a professor). With the SP4 and one omni at 0.75 m: **SRI 70 m +36.5 dB, MRM 260 m +13.9 dB** (+8.4 dB at today's RF_PWR −12 until T0 measures the module). NM stays at 10 mW on one channel (UN-85 b): **+9.9 dB**.
3. **MRM is no longer the thin link.** +13.9 dB at 0.75 m, +6.9 dB even at 0.5 m. All teams transmit at the same 20 dBm, so the near-far ratios are back to the equal-power ones:

keep **≥ 10 m** between teams' boxes at each end (never 2 m).

4. **Speed is no longer a problem.** 250 kbps gives ~107 kbps of goodput (~96 kbps hopping). The simulator delivers the whole file in 27 s at 5 % loss, and all 10,000 lines even without compression (71 s).
5. **One omni, decided.** Plugged straight into the SP4 (no adapter) and with the SP4's −96 dBm sensitivity, it is only 1 dB behind the two-patch build on MRM, with none of the patches' NM risks. Remaining risk: if T0 measures only ~+2 dBm out of the module, every link loses ~5 dB.
6. **MRM channel plan:** none any more. Every team hops at 20 dBm with listen-before-talk; ours uses 37 channels in its own order. Hops that land within 4 MHz of another team's collide: about 13 % of hop pairs at 10–20 m, which costs up to ~31 % of our ACKs at the sender end (pessimistic). The ARQ absorbs it, with a short POLL for lost ACKs: all lines in 58 s in the simulator (115 s with 30 % loss both ways).
7. **Plan for less.** With the datasheet sensitivity 3 dB worse, the antenna's worst direction (0 dBi) at both ends and a 6 dB body or fade loss, MRM at 20 dBm keeps **+0.9 dB** and NM falls to **−3.1 dB**. Antenna height, people kept away from the boxes, and tests T4/T5 decide (link budget).

Working baseline decided

The **E01-ML01SP4** is a genuine Nordic nRF24L01+ with a PA and an LNA, as a 14.85 × 18 mm SMD module with an **IPEX (U.FL) antenna socket**. Datasheet: +20 dBm max, **−96 dBm at 250 kbps**, 120 mA TX, 26 mA RX, 2.0–3.6 V `VERIFIED`. Its lowest power setting gives about **+2 to +7 dBm**, depending on the PA gain; test T0 measures it before any field test `ASSUMPTION`. With the 3 dBi peak of the omni that is at most 9.5 dBm e.i.r.p. That is the 10 mW setting for NM, Quick Mode and the fallback. The hopping profile for SRI and MRM may go up to 20 dBm e.i.r.p. (below); the driver refuses anything higher on every other profile.

Four radio profiles are generated by `radio/phy_config.py`. Each is checked against the band edges (Carson bandwidth plus ±60 ppm crystal drift per box) and the 10 dBm e.i.r.p. limit for its antenna path.

PROFILE	USE	CHANNEL	RATE	AT SMA	ANTENNA I
TEAM_250K	Quick Mode, bench,	74 · 2474 MHz	250 kbps	≤ +7 dBm	omni 3 dB peak, −0.5

PROFILE	USE	CHANNEL	RATE	AT SMA	ANTENNA I
	fallback (10 mW)				
TEAM_1M	Fallback option at 1 Mbps	74 · 2474 MHz	1 Mbps	≤ +7 dBm	omni 3 dB peak, −0.5
NM_COMMON	Shared with all teams (NM spec)	77 · 2477 MHz	250 kbps	≤ +7 dBm	omni 3 dB peak, −0.5
TEAM_FHSS	**SRI and MRM baseline:** hopping with listen-before-talk (D-R6)	37 hops · 2404–2478	250 kbps	≤ +13 dBm (−12)	omni 3 dB peak, −0.5
TEAM_20DBM_FIXED	Option B: SRI/MRM on one fixed channel at 20 dBm (only on an explicit yes)	74 · 2474 MHz	250 kbps	≤ +13 dBm (−12)	omni 3 dB peak, −0.5

No duty-cycle limit

Neither UN-85 b) nor adaptive hopping under UN-85 a) has a duty-cycle condition, so the 3-rounds-per-hour limit of 868 MHz is gone. Test as often as we like. The firmware's duty-cycle guard was removed.

Firmware ready before the hardware

A portable C driver for the nRF24L01+ passes 318 host checks against a simulated radio, including two radios talking over a simulated channel, the power limit, and the listen-before-talk rules of the hopping profile. It

The power limit is the only rule to watch. On the 10 mW profiles the driver **refuses any setting above RF_PWR −18 dBm**; only the hopping profile goes higher, and only after a clear-channel check on each hop.

needs five board functions (SPI select and transfer, CE pin, microsecond delay and clock).

Frames use dynamic length and the radio's CRC, and go out without the chip's own ACKs, so our ARQ stays in charge. The protocol simulator runs on it: all 21 scenarios pass, six of them hopping (T19 adds a 30 s Linux reboot).

Antenna

One Molex 146153-0100 flex dipole per box (34.9 × 9 mm, 2.4–3.0 dBi peak, > 66 % efficiency), plugged into the SP4's socket, stuck **vertically** as high as possible on a side wall. The same antenna serves SRI, MRM and NM, and Quick Mode too.

The model uses 2 dBi in the box `ASSUMPTION`; test T5 measures it.

What to buy (order by Fri 16 Oct)

ITEM	QTY	PRICE	WHY
Ebyte E01-ML01SP4 (nRF24L01P + PA/LNA, SMD, IPEX)	6	verify	2 boxes, 2 for the bench, 2 spares. Modules shown at Quick Mode are final, so buy them all from one batch, and check the EU seller and lead time first.
Molex 146153-0100 flex omni, 100 mm cable, U.FL	4	≈ €2–4	The antenna. Check the part is the MHF (U.FL) version, not MHF4.
U.FL-to-SMA-female pigtail	2	€2–3	To measure the module on the spectrum analyser (T0, T13).

Do not buy Ebyte E01C-* parts or no-name "nRF24L01+PA+LNA" boards: Ebyte's own manual says the E01C uses the Si24R1 clone chip. The SP4 is SMD, so it needs a small breakout (or the carrier PCB) from the first bench test. Not needed any more: the DP5, SMA pads and whips.

Antenna: one omni decided 13 Oct

The antennas cannot change between SRI, MRM and NM, so one build has to serve all three. The comparison we decided on, model at 0.75 m (the first row is the decision; the others are the 9 Oct options):

BUILD	E.I.R.P.	SRI 70 M	MRM 260 M	MRM AT 1.0 M	NM 100 M, −10 DB
SP4 + one omni, plugged straight in (decided), 10 mW one channel (NM; SRI/MRM fallback)	8.5 dBm	+26.0	+3.4	+8.3	+9.9
DP5 + two patches, 4.5 dBi (not chosen)	10.0 dBm	+27.0	+4.4	+9.3	+10.9
DP5 + one omni 2 dBi, via adapter	8.0 dBm	+23.0	+0.4	+5.3	+6.9
One omni straight on the SMA	8.7 dBm	+24.4	+1.8	+6.7	+8.3
One omni, 0 dBi inside the box	6.0 dBm	+19.0	−3.6	+1.3	+2.9
SP4 + one omni + 20 dBm hopping: SRI/MRM baseline	19.0 dBm	+36.5	+13.9	+18.8	+9.9 NM stays at 10 mW

Why one omni

- Covers every direction in NM, including the box at the corner of the two rims, with no aiming
- One radio per box, no patch boards, no VNA tuning, off-the-shelf antenna
- Only ~1 dB behind the patches on MRM once it plugs straight into the SP4

What we give up

- At 10 mW the MRM margin was thin (+3.4 dB at 0.75 m); hopping at 20 dBm lifts it to +13.9 dB. Mounting the antenna high still adds +5 dB per 25 cm
- No sideways rejection of other teams' boxes: the ≥ 10 m spacing of the channel plan still holds, with no extra isolation

Why not the E01-2G4M27D

Its −99 dBm receiver is tempting, but its lowest output is about +9 dBm. Behind a ~3 dBi antenna that is ~12 dBm e.i.r.p., over the

10 mW limit. It would be legal in the 20 dBm hopping mode, but it could not run NM, Quick Mode or the fallback at 10 mW, so it stays out.

20 dBm with frequency hopping baseline · 16 Oct

CNAF UN-85 has two regimes in 2400–2483.5 MHz `VERIFIED`. **b)** generic low-power devices: under 10 mW e.i.r.p.: NM, Quick Mode and the fallback. **a)** wideband data systems under EN 300 328: **100 mW e.i.r.p.**, but the density is capped at **100 mW per 100 kHz for frequency hopping** and **10 mW per MHz for anything else**. Our signal is under 1 MHz wide, so without hopping a) still means ~10 dBm. The 20 dBm needs a hopping radio that follows the EN 300 328 sharing rules.

Without hopping, EN 300 328 also allows 20 dBm in total, but caps the density at **10 dBm per MHz** (cl. 4.3.2.3), as does the EU decision that UN-85 a) points to. Our signal is about 0.6 MHz wide, so on one fixed channel it is 5–10 dB over (option B below). EN 300 328 V2.2.2 has two kinds of hopping. **Non-adaptive** hopping caps power × duty: at 20 dBm the radio may transmit only 10 % of the time (~16 kbps), too slow for a 2-minute file. **Adaptive hopping with listen-before-talk** has no duty limit, so that is what SRI and MRM use (D-R6). A professor approved 20 dBm e.i.r.p. on 16 Oct.

EN 300 328 RULE (ADAPTIVE FHSS, LBT)	REQUIREMENT	OUR DESIGN (PHY_CONFIG.PY, ALL CHECKS OK)
RF output power (4.3.1.2)	≤ 20 dBm e.i.r.p.	RF_PWR −12 today: ≤ 15.5 dBm at the worst-case PA gain. If T0 measures the gain at ≤ 21.5 dB, −6 is legal: up to 20 dBm.
Hop channels (4.3.1.4)	≥ 15 at all times, over ≥ 70 % of the band	**37 channels**, the even RF_CH 4–78 except 26 (BLE advertising channels 37/38/39 left out), about 75 MHz = 90 % of the band. Our own order (stride 7); other teams should pick another
Time per channel (4.3.1.4)	≤ 400 ms in any 400 ms × N window	30 ms per hop (one 16-frame burst plus its ACK), each channel once per 1.1 s cycle: ≤ 150 ms
Hop separation (4.3.1.5)	≥ 100 kHz	2 MHz between neighbours; consecutive hops ≥ 14 MHz apart

EN 300 328 RULE (ADAPTIVE FHSS, LBT)	REQUIREMENT	OUR DESIGN (`PHY_CONFIG.PY`, ALL CHECKS OK)
Clear channel check (4.3.1.7.2)	Energy detect ≥ 18 µs before each hop's first frame; ≤ −70 dBm/MHz at 20 dBm (relaxed at lower power)	The nRF24's RPD bit after 170 µs in RX: ~−76 dBm at the SP4 port `ESTIMATE`, needed ≤ −64 dBm at −12 and ≤ −68.5 dBm at 20 dBm. **T17 measures it.**
Channel occupancy (4.3.1.7.2)	< 60 ms per check, then idle ≥ 5 %	50 ms, then 2.5 ms idle; busy hops are skipped

Two versions of 20 dBm

	OPTION A: HOPPING (DEFAULT)	OPTION B: ONE FIXED CHANNEL
Profile	`TEAM_FHSS: 37 hops, 30 ms each`	`TEAM_20DBM_FIXED: RF_CH 74, MRM plan 24 / 49 / 74 / 82`
EN 300 328 class	adaptive FHSS with LBT (4.3.1.7.2)	adaptive non-FHSS, frame-based LBT (4.3.2.6.3.2.2)
Listen-before-talk	a check per hop, ≤ 50 ms per check	a check every ≤ 10 ms (about 6 frames), 0.5 ms idle
Power density	**compliant: 100 mW/100 kHz allowed**	**5–10 dB over 10 dBm/MHz**
Margins (SRI 70 m · MRM 260 m)	+36.5 · +13.9 dB	same
Firmware	hop clock and acquisition on the Pi 3	none extra: today's link on one channel
MRM with the other teams	random hop collisions, ≥ 10 m between boxes	fixed plan at equal power: the 8 MHz pair is fine at ≥ 10 m
Use it	**default**	**only on an explicit yes (Atenea Q11)**

Sources: ETSI EN 300 328 V2.2.2 cl. 4.3.1 and 4.3.2; CNAF UN-85 (2013/2017 text) and Commission Decision 2006/771/EC as amended (2019/1345, 2022/180). The CNAF in force since July 2026 is Orden TDF/732/2026 (BOE-A-2026-15661); its consolidated text does not reproduce the UN notes, so the EU decision is the reference. Switching between A and B is one firmware profile.

What changes

- **Nothing in the hardware.** The SP4 is a +20 dBm module that we hold at its lowest step; hopping only raises the step. TX bursts move toward the datasheet's 120 mA, which the radio LDO already covers.
- **Driver:** only the hopping profile may go above −18 dBm. It refuses channels outside the hop set and any frame without a clear check on the current hop, ends each occupancy at 50 ms and waits the idle gap. 318 host checks pass.
- **Protocol:** unchanged; the radio layer hops every 30 ms (one 16-frame burst). The simulator gives each box its own drifting slot clock; the receiver parks on one channel and re-syncs. All 10,000 lines in 38 s with 10 % of hops busy, 69 s with 25 % busy plus a receiver reboot. Still to build on the Pi 3: the slot clock and hop acquisition (only simulated so far); IRQ edges with kernel timestamps make the acquisition accurate under Linux.
- **NM stays on one channel (77) at 10 mW.** Hopping in sync across four teams' boxes for a 0.5 kB file is not worth it, and NM has +9.9 dB.

Why it is allowed, and what to keep

- **Regulation:** the design meets the EN 300 328 adaptive-FHSS rules in the table, at ≤ 20 dBm e.i.r.p. measured with our antenna.
- **Course:** a professor approved 20 dBm e.i.r.p. on 16 Oct, and the answer to **Atenea Q3** confirmed it in writing (the rules say oral comments do not override the written rules). Cite Q3 in the ANNEX.
- **Evidence:** T13 (e.i.r.p.), T17 (where the clear-channel check trips) and T18 (hop pattern, dwell, skipping a busy channel on the analyser), plotted for the ANNEX.
- **Other teams:** they use 20 dBm too, so they hop with listen-before-talk as well. Agree **≥ 10 m between boxes** at each MRM end (see the MRM section).

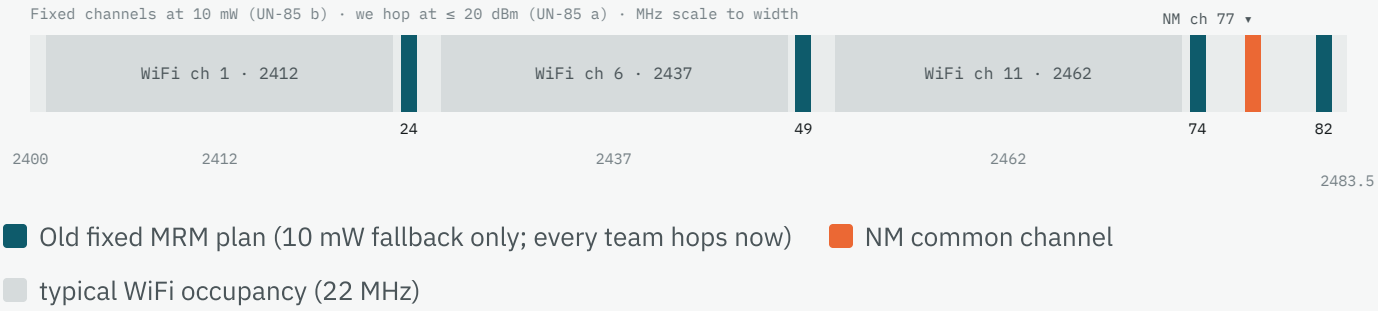
MODEL AT 0.75 M (WORKBOOK S24–S26)	E.I.R.P.	SRI 70 M	MRM 260 M	MRM AT 0.5 M	MRM AT 1.0
10 mW, one channel (fallback)	8.5 dBm	+26.0	+3.4	−3.6	+8.3
Hopping at 20 dBm, RF_PWR −6 (baseline once T0 allows)	19.0 dBm	+36.5	+13.9	+6.9	+18.8
Hopping at RF_PWR −12 (today, until T0)	13.5 dBm	+31.0	+8.4	+1.4	+13.3

Rules that constrain the radio

- Unlicensed spectrum only, compliant with **CNAF 2017 UN notes**, IEEE C95.1 (RF exposure) and ESD safety. RULE
- **No chip implementing a standard wireless access protocol** (WiFi, Bluetooth, 2G–5G, ZigBee; the Session 1 slide adds NB-IoT and **LoRa**). The nRF24L01+ has none of them. RULE
- **Nothing may stick out of the 15 × 15 × 7 cm IKEA 365+ box** except buttons, switches and connectors, so the antennas go inside. The box sits on a support of **about 70 cm**. RULE
- **Modules shown at Quick Mode are final.** Antennas, batteries and mechanics may still change. RULE
- Distances: SRI **70 m**, teams one after another, 2 minutes; MRM **~260 m**, **all teams at the same time**, 2 minutes; NM relays along the South and East rims with no line of sight end to end, **5 minutes**. RULE
- Scoring counts only the **first unbroken run of byte-identical lines** (~10,000 lines of ~100 B), so reliability beats raw bit rate. RULE

Spectrum: 2400–2483.5 MHz and our channels

nRF24 channel N sits at 2400 + N MHz, so only **RF_CH 0–83** are legal in Spain; the driver refuses anything higher. Campus WiFi (eduroam) normally uses channels 1, 6 and 11, which leave three gaps. The old fixed MRM plan (now only the 10 mW fallback) sits in those gaps. Since 16 Oct every team hops at 20 dBm across the band; ours uses 37 even channels from 2404 to 2478 MHz (not the BLE advertising channels 2402/2426/2480), and listen-before-talk skips whatever is busy. Test T14 (campus survey) confirms it before we propose it.



BAND	NOTE	MAX POWER	ACCESS CONDITION	STATUS FOR US	COMMENT
2400–2483.5 MHz	UN-85 b)	10 mW e.i.r.p.	none (EN 300 440)	NM · Quick Mode · fallback	No duty cycle. Eve team gets

BAND	NOTE	MAX POWER	ACCESS CONDITION	STATUS FOR US	COMMENT
					own char in MRM. Shared w campus V and Bluetooth
2400–2483.5 MHz	UN-85 a)	100 mW e.i.r.p.; 100 mW/100 kHz hopping, 10 mW/MHz other	sharing rules required (EN 300 328)	SRI · MRM baseline	A non-hopping narrowba radio gair nothing (1 mW/MHz Our SRI/MRM hopping mode wit listen-before-ta uses it sir 16 Oct (a professor approved dBm): +1 dB. NM st under b).
869.40–869.65 MHz	UN-39	500 mW e.r.p.	≤10 % DC	dropped 9 Oct	Better ran but one 2 kHz chan for four simultane teams in MRM and rounds pe hour.
433 / 5725 MHz	UN-30 / UN-130	10 / 25 mW	≤10 % DC / none	not used	433 MHz antenna t long for th box; few cheap no

BAND	NOTE	MAX POWER	ACCESS CONDITION	STATUS FOR US	COMMENT
					WiFi part 5.8 GHz.

Sources: CNAF 2017, BOE-A-2017-12318 · CNAF 2021, BOE-A-2021-21346.

MRM: every team hops at 20 dBm

Our ARQ makes both ends transmit (data one way, ACKs back), and the four teams' boxes stand close together at each end. Another team's box a few metres away is far louder than our partner 260 m away. Whether we still hear our partner depends on how far apart the channels are. The nRF24's selectivity at 250 kbps is **−50 dB beyond 6 MHz** and **−60 dB beyond 25 MHz** (datasheet, table 8) `VERIFIED`.

OTHER TEAM'S BOX FROM OUR RECEIVER	2 M	5 M	10 M	20 M
Interferer above our 260 m signal, omni antennas (our build)	55 dB	47 dB	41 dB	35 dB
Same with patches side by side (~10 dB rejection each; not our build)	35 dB	27 dB	21 dB	15 dB

CHANNEL PAIR	SPACING	SELECTIVITY	WHAT IT NEEDS
24 / 49 · 2424 / 2449 MHz	25 MHz	−60 dB	safe even at 2 m
49 / 74 · 2449 / 2474 MHz	25 MHz	−60 dB	safe even at 2 m
74 / 82 · 2474 / 2482 MHz	8 MHz	−50 dB	fallback plan only

SINCE 16 OCT: ALL TEAMS HOP AT 20 DBM. ANOTHER TEAM'S BOX R M AWAY	2 M	5 M	10
It lands above our 260 m partner (same power everywhere), +6 dB two-ray peak	61.4 dB	53.4 dB	47.4 d
Hop offsets that hurt (selectivity too small)	all	≤ 24 MHz	≤ 4 MH
Hop pairs that collide (37 channels each)	100 %	55 %	13 %
Lost ACKs at our sender (3 senders next to it, ~90 % busy; no CCA credit)	~100 %	87 %	31 %

SINCE 16 OCT: ALL TEAMS HOP AT 20 DBM. ANOTHER TEAM'S BOX R M AWAY	2 M	5 M	10
Lost DATA at our receiver (3 receivers next to it send ACKs ~5 % of the time)	14 %	8 %	2 %

Pessimistic: it ignores the clear-channel check, which skips a hop when another team is on the same channel nearby. At 30 m the collisions fall to 8 % of hop pairs (20 % of ACKs). At any spacing, ~7 % of our ACKs meet another sender on the very same channel, which the clear-channel check cannot see while we listen. Hence the POLL: after a lost ACK the sender asks for it again with one 2-byte frame instead of resending 16. Simulator T14 (5 % data loss, 30 % ACK loss, 10 % busy hops): all 10,000 lines in 58 s; with 30 % loss both ways, 115 s. Agree **≥ 10 m between teams' boxes** at each end (**30 m between the senders** if the organisers allow it), a different hop order per team, and listen-before-talk for everyone. Test T15 checks it with a third box hopping 10 m away.

Link budget

This uses the course method: Maxim AN5142 with the course correction (no ½ factor in the flat-earth formula) and the course spreadsheet's worst-case ground bounce (a = 1, |cos|). Inputs:

- antennas at 0.75 m (70 cm support plus position in the box) ASSUMPTION
- −1 dB container loss and 0 dB multipath on open campus paths ASSUMPTION
- flex omni 2 dBi in the box, cable −0.5 dB, in both directions; +7 dBm out of the module at 10 mW, up to +17.5 dBm while hopping ASSUMPTION
- receiver sensitivity from the E01-ML01SP4 datasheet (−96 dBm at 250 kbps) VERIFIED

FLAT-EARTH LINK MARGIN VS DISTANCE · DB ABOVE RECEIVER SENSITIVITY

SP4 + omni · 20 dBm hopping (SRI/MRM baseline) SP4 + omni · 10 mW, one channel (NM, fallback)

SP4 + omni · 1 Mbps, 10 mW RFM69 868 MHz · 100 kbps (dropped)



▶ Show as table

SCENARIO	EIRP DBM	RX DBM	SENS. DBM	MARGIN DB	WITH −20 DB MULTIPATH
SRI 70 m · hopping at 20 dBm (baseline)	19.0	-59.5	-96.0	+36.5	+16.4
MRM 260 m · hopping at 20 dBm (baseline)	19.0	-82.1	-96.0	+13.9	-6.1
MRM 260 m · hopping at RF_PWR −12 (today, until T0)	13.5	-87.6	-96.0	+8.4	-11.6
NM 100 m hop, −10 dB obstruction · 10 mW, one channel (baseline)	8.5	-86.1	-96.0	+9.9	-10.1
SRI 70 m · 10 mW, one channel (Quick Mode, fallback)	8.5	-70.0	-96.0	+25.9	+6.0
MRM 260 m · 10 mW, one channel (fallback)	8.5	-92.6	-96.0	+3.4	-16.6

Conservative planning case (16-Oct review): the receiver 3 dB worse than the datasheet (for 1 % packet loss), the omni's worst direction (0 dBi) at both ends, and −6 dB for fading or a person near the box.

- SRI at 20 dBm: +36.5 → **+23.5 dB**.
- MRM at 20 dBm: +13.9 → **+0.9 dB**; at today's RF_PWR −12: **−4.6 dB**.

- NM at 10 mW: +9.9 → **−3.1 dB**.

So MRM and NM are not safe on paper. The levers, in order:

- mount the antenna as high as the box allows (+5 dB per 25 cm);
- keep people away from the boxes;
- use the highest legal power step after T0 (−6 if the PA gain is ≤ 21.5 dB);
- measure in T4/T5: attenuate step by step until the link fails, so we know the real margin.

Full AN5142 rows are in `radio/MTP-F26_TeamC_Radio_LinkBudget.xlsx` (S24–S26 are the SRI/MRM baseline at 20 dBm hopping, S23 is NM at 10 mW, S21–S22 the 10 mW fallback; S16–S20 the patch option); every cell is a live formula and inputs are in blue. AN5142's default −20 dB multipath is for cluttered paths. Tests T4 and T5 on campus will tell us which number to use.

Antenna height is the biggest lever

At 0.75 m the ground reflection takes over beyond ~18 m at 2.4 GHz. Past that point loss grows 40 dB per decade and hardly depends on frequency: $P_R \approx EIRP \cdot G_R \cdot (h_1 h_2)^2 / d^4$.

Raising both antennas from 0.75 to 1.0 m adds **+5 dB** on MRM. Mount the omni as high on the wall as the box allows. Whether "about 70 cm" allows a taller support is Atenea Q4.

The nRF24 has no RSSI

It only has one bit (RPD) that says "above −64 dBm at the chip", about −76 dBm at the SP4's antenna socket (LNA gain to be measured). Range tests therefore measure packet loss, and levels come from the spectrum analyser or the model. RPD still works as listen-before-talk and for the campus channel survey.

Datasheet limits that matter

- **32 B maximum payload:** our frames shrank from 64 B to 32 B (28 B of data). `VERIFIED`
- **Variable-length frames need the chip's auto-ACK feature on,** so the driver sends every frame as "no ACK": no chip ACKs or retries on air.
- **PA always in line:** no bypass, so the lowest output is +2 to +7 dBm; the driver caps the setting there.

Throughput: does the file fit in 2 minutes?

About 1 MB in 120 s needs **66.7 kbps** of useful throughput. The model assumes selective-repeat ARQ with one ACK per 16 frames, 28 B of file per 32 B frame, a 0.3 ms gap between frames, and 85 % efficiency for the micro and USB `ASSUMPTION`. The simulator runs the real protocol code over a modelled channel with 5 % random loss, deflate compression, and the nRF24 airtime from the driver.

CONFIGURATION	AIR RATE	GOODPUT (MODEL)	WHOLE FILE (SIM, 5 % LOSS)	LINES, NO COMPRESSION
SP4 hopping, 32 B frames, 10 % of hops busy (MRM baseline)	250 kbps	95.7 kbps	38 s	10,000
SP4, 32 B frames, one fixed channel (SRI, Quick Mode)	250 kbps	107.4 kbps	27 s	10,000 (71 s)
SP4, 32 B frames, one fixed channel	1 Mbps	249.8 kbps	12 s	10,000
RFM69 868 MHz, 64 B frames (dropped)	100 kbps	67.3 kbps	38 s	9,643
RFM69 868 MHz shared by 4 teams in MRM (dropped)	~25 kbps	~17 kbps	—	~2,500

> At 2.4 GHz each team keeps most of its rate in MRM, because the teams hop across 83.5 MHz instead of splitting one 250 kHz channel. That, not range, is why the band switch pays off.

Module shortlist

Read from the primary datasheets on 9 Oct 2026. Prices and EU lead times are still **TO VERIFY**.

MODULE	CHIP	OUT AT THE CONNECTOR	SENSITIVITY	CONNECTOR	PROTOC
Ebyte E01-ML01SP4 decided	nRF24L01P + PA/LNA	~+2 to +20 dBm	−96 @250k (datasheet)	IPEX on the module	low–m
Ebyte E01-ML01DP5 not chosen	nRF24L01P + PA/LNA	~+2 to +20 dBm	~−96 @250k	SMA	low–m
Ebyte E01-2G4M27D	nRF24L01P + 27 dBm PA	≥ +9 dBm	−99 @250k	SMA	low–m
Bare nRF24L01+	nRF24L01+	−18 to 0 dBm	−94 @250k	PCB / none	low–m
TI CC2500 modules	CC2500	+1 dBm	−87 @250k	PCB	low
E01C-*, no-name	Si24R1 clone	unknown		SMA	exclud

MODULE	CHIP	OUT AT THE CONNECTOR	SENSITIVITY	CONNECTOR	PROTOC
PA+LNA boards					
SX1280 / SX1281	LoRa 2.4 GHz	—			exclud
Radio SoCs (nRF52, ESP32, CC2652…)	—	—			exclud

Decisions and who they touch

DECISION	OWNER	DEPENDS ON	AFFECTS	STATUS
D-R1b Band: 2.4 GHz for every mode, 250 kbps	All teams + Radio	—	Everything radio	agreed by the teams, 9 Oct
D-R2c Radio module (frozen at Quick Mode)	Radio → handed to electronics	T0 output power, T2 sensitivity	PCB, BoM, power supply	decided 13 Oct: E01-ML01SP4 order by Fri 16 Oct
D-R3 NM radio and protocol shared by all teams	Radio + protocols	Other teams' modules	NM bonus (+2 to +2.5)	spec v1.0 draft ready agree by 23 Oct; interop test by 6 Nov
D-R4 MRM channel per team	Radio + other teams	Campus survey T14	MRM (+1)	superseded: every team hops fixed plan = 10 mW fallback
D-R6 SRI/MRM as adaptive hopping + listen-before-talk at ≤ 20 dBm e.i.r.p.	Radio + protocols	Atenea Q3 = yes; T0 (all steps), T17, T18	MRM (+10.5 dB), firmware, other teams	baseline 16 Oct built and simulated; bench tests next

DECISION	OWNER	DEPENDS ON	AFFECTS	STATUS
D-R6b Option B: 20 dBm on one fixed channel, no hopping	Radio	Atenea Q11	Firmware (simpler), MRM plan	**built, not in use** only on an explicit yes
D-R5c Antenna: one flex omni (Molex 146153-0100)	Radio	Box inside height, T5	Mechanics, carrier PCB	**decided 13 Oct** T5 measures the in-box gain

> **Revisit D-R2c if** T0 measures more than ~+7.5 dBm at the lowest setting (the e.i.r.p. limit with a 3 dBi antenna), or T2 shows a clone or a weak receiver. If T0 measures only ~+2 dBm, the module is legal but every margin drops ~5 dB: then raise the antenna and run the hopping profile at the highest legal step (D-R6).

Handoff to electronics

One **E01-ML01SP4** per box. SMD, 10 pads at 1.27 mm: 1 VCC, 2 CE, 3 CSN, 4 SCK, 5 MOSI, 6 MISO, 7 IRQ, 8–10 GND (Ebyte manual §3). No traces under the module on the top layer.

PIN	AT THE MICRO	WHAT MATTERS
`VCC, GND`	supply	**2.0–3.6 V only**, never the battery rail (a Li-ion cell reaches 4.2 V). Use a dedicated 3.3 V LDO for the radio (≥ 300 mA, dropout ≤ 0.1 V at 150 mA) straight from the cell: it regulates down to ~3.4 V and then passes the cell through (≥ 2.95 V at cutoff, fine for the SP4). Never the Pi's 3V3 pin. Battery sense with a MAX17048 fuel gauge on I2C (the Pi has no ADC); firmware warns below 3.5 V, and 100 µF + 10 µF + 100 nF at the module. 120 mA in TX, 26 mA in RX.
`SCK MOSI MISO`	SPI master	Mode 0, MSB first, ≤ 10 MHz (on the Pi: 7.8 MHz with `core_freq=250`).
`CSN`	GPIO out	Chip select, active low. On the Pi it is SPI0 CE0, driven by the kernel for each transfer.
`CE`	GPIO out	High = receive; a ≥ 10 µs pulse sends one frame.

PIN	AT THE MICRO	WHAT MATTERS
IRQ	GPIO in, optional	Active low: frame sent / frame received. The driver polls today; on the Pi, wire it, because edge timestamps help the hop sync.
IPEX	—	The flex omni plugs in here. Keep the module near the antenna wall and away from the battery, USB, the 5 V boost and the Pi's processor.

Proposed Pi 3 pins (BCM numbers): SPI0 SCK GPIO 11, MOSI GPIO 10, MISO GPIO 9, CSN = CE0 GPIO 8, CE GPIO 22, IRQ GPIO 24; full table on the Pi 3 Switch page. The SP4 needs a small breakout (or the carrier PCB) from the first bench test, because the Quick Mode module is final. Needed back from electronics: OK on the module and the radio LDO, who makes the breakout, the pin map, a power check, and the box's inside height.

Shared NM radio settings: draft for all teams

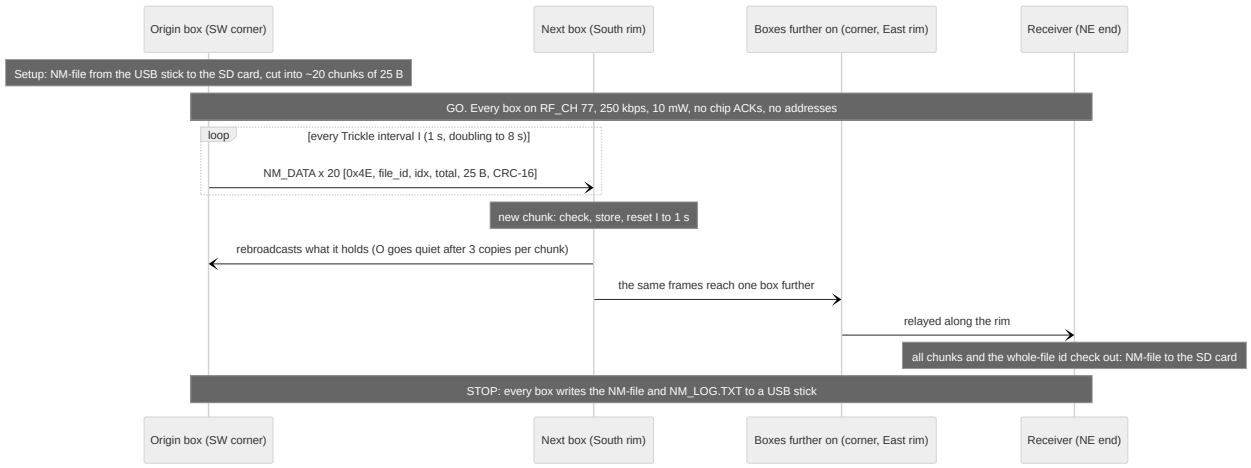
Plain nRF24L01+ register settings, so any team's nRF24 board can join with a few register writes and no shared code. **The full spec to share with the other teams** (registers in hex, frame, gossip rules, CRC, test vector): NM Interop Spec v1.0, source `radio/NM_SPEC.md`.

SETTING	VALUE	WHY
Channel	`RF_CH 77 · 2477 MHz`	Above WiFi 11 and 3 MHz from BLE advertising channel 39 (2480); NM rounds run alone, so the MRM hopping doesn't meet it
Rate	`250 kbps`	Best sensitivity and selectivity; NM moves only ~0.5 kB
Address	`5 B · C2 5A 9D 3E 71`	Bytes in SPI order (0x713E9D5AC2 as a 40-bit number); different from every team's own address
Frames	dynamic length 1–32 B, hardware CRC-16	Same on every nRF24L01+
ACKs	none on air: send with "no ACK", retries 0	Chip ACKs from several receivers would collide
Power · polarisation	8–10 dBm e.i.r.p. · vertical	Legal limit, and a floor so the weakest box still reaches the next; mismatched polarisation costs 10–20 dB

SETTING	VALUE	WHY
Channel access	listen before talk with RPD, random back-off	The nRF24 has no carrier sense in hardware

Interop test by Fri 6 Nov: every pair of boxes exchanges 1 000 frames at 10 m on RF_CH 77, with a pass at ≥ 99 % with a valid end-to-end CRC, then a gossip run with the spec's test-vector file; full rehearsal Fri 20 Nov.

NM link protocol (gossip, no routing)



Every box ends up with an identical copy, which is how a team proves it took part (rules: "delivering a copy of the NM-file from at least one of their node devices"). The relay log is extra evidence. Details: the protocol proposal page.

Test plan

No duty-cycle limit, so tests can be repeated freely. Stay at RF_PWR −18 dBm until T0; the hopping profile goes outdoors once T17 and T18 pass in the lab. For each run, record place, profile, channel, distance, heights, antenna, box open or closed, pad, and frames sent and received.

TEST	WHAT	PASS	WHEN
T0	Bring-up, then **conducted output power at all four settings** on the spectrum analyser (U.FL-to-SMA	Driver starts; output ≤ +7.5 dBm at −18, so e.i.r.p. ≤ 10 dBm with the 3 dBi antenna. The other steps pick the hopping mode's power. **Gate for any field test.** With the Pi 3: also log RPD hits and packet loss on every hop channel with the Pi idle and busy (clock	modules arrive

TEST	WHAT	PASS	WHEN
	pigtail, ≥ 30 dB attenuator)	harmonics, 6 × 400 MHz = 2400 MHz), and the Pi's current at 3.6 / 3.3 / 3.0 V.	
T2	Sensitivity at 250 kbps and 1 Mbps through a lab step attenuator	1 % packet loss at ≤ **−92 dBm** (250 kbps). **Gate for the module.** On the Pi, also measure the gap between frames and the TX↔RX turnaround (model: 0.3 ms, 1 ms).	before Quick Mode
T14	Campus 2.4 GHz survey with the RPD bit, channels 0–83, at each site	Our hop channels and the NM channel (77) busy < 5 %; any team's hopping visible	before the inter-team meeting
T4	Range, open box, 10 / 70 / 150 / 260 m; then one 2-min transfer at 70 m	≤ 1 % loss to 150 m; all 10,000 lines	after T2
T13	Spectrum analyser: carrier offset, occupied bandwidth on channel 82, e.i.r.p. (option B: PSD in 1 MHz)	Below 2483.5 MHz; ≤ 10 dBm (≤ 20 dBm hopping). Plot in the ANNEX.	book the lab
T17	D-R6: input level where the clear-channel check (RPD) trips, from a signal generator	≤ the threshold for our power (−64 dBm at −12, −68.5 dBm at 20 dBm)	with T13
T18	D-R6: hop pattern, dwell and occupancy on the analyser; skips a channel with a CW interferer on it	37 channels, ≤ 30 ms dwell, ≤ 50 ms per check, busy channel skipped	with T13
T5	Closed box; antenna high vs middle of the wall; box rotated 0 / 90 / 180°	≤ 1 % loss at 260 m	after T4
T15	MRM near-far with a third box hopping at 20 dBm, 10 m from our receiver and then from our sender	≤ 1 % loss on the 260 m link	after T5

EXPECTED (MODEL)	10 M	70 M	150 M	260 M
SP4 + omni · level	−56 dBm	−70 dBm	−83 dBm	−93 dBm
SP4 + omni · margin	+40 dB	+26 dB	+13 dB	+3 dB

Next: to Fri 23 Oct

ITEM	DATE ASG.	ASSIGNED TO	DESCRIPTION	STATUS	DUE DATE	END DATE	RESULT
R7	09/10/26	RT + PM	Band, module and antenna decisions	Done	16/10/26	13/10/26	2.4 GHz (09/10); SP4 + one omni (13/10)
R12	13/10/26	RT → ET	Order 6 × E01-ML01SP4, 4 × Molex 146153-0100, 2 × U.FL-to-SMA pigtails	Open	16/10/26		
R13	09/10/26	Andrian	Campus 2.4 GHz survey (T14) at the SRI, MRM and NM sites	Open	16/10/26		
R14	09/10/26	RT	MRM channel plan + NM radio settings v0.2 to the other teams' leaders	Open	16/10/26		
R9	13/10/26	RT → ET	Electronics handoff v0.3 reviewed; pin map, radio LDO, SP4 breakout and box height back	Open	16/10/26		

ITEM	DATE ASG.	ASSIGNED TO	DESCRIPTION	STATUS	DUE DATE	END DATE	RESULT
R15	16/10/26	Marcos	Post Atenea Q3: written confirmation of the 20 dBm the professor approved	Done	23/10/26	16/10/26	Yes: 20 dBm e.i.r.p. confirmed for our hopping design
R16	16/10/26	RT → teams	Agree with the other teams: everyone hops with listen-before-talk at ≤ 20 dBm in its own order; ≥ 10 m between boxes at each MRM end; the NM interop spec (RF_CH 77, 10 mW)	Open	23/10/26		
R11	04/10/26	RT	Tests T0 and T2 done; D-R2c (E01-ML01SP4) confirmed or reversed	Open	before QM		
R17	18/10/26	Marcos + ET	Raspberry Pi 3 as the micro: post Atenea Q22 (WiFi/BT chip disabled or desoldered); ET picks the model (A+ suggested) and counts the boards we have; Pico 2s dropped from the order	Open	23/10/26		

Questions to ask

Atenea forum (posted by the project leader)

- **Allowed chips.** Is the nRF24L01+ on the Ebyte E01-ML01SP4 allowed? Its packet format is Nordic-proprietary and it has no Bluetooth.
- **20 dBm with hopping (Q3): answered yes, 16 Oct.** 20 dBm e.i.r.p. is confirmed for our SRI/MRM hopping design; nothing left to ask.
- **20 dBm without hopping (Q11, new).** May we use the confirmed 20 dBm on one fixed channel (about 0.6 MHz wide) with listen-before-talk every ≤ 10 ms? EN 300 328 limits non-hopping radios to 10 mW/MHz, which such a narrow signal exceeds by 5–10 dB. If not, we keep hopping.
- **Support height and box.** Is the support fixed at 70 cm? May the box stand on its 15 × 7 cm edge?
- **Cooperation.** May teams agree a common NM radio and an MRM channel split, sharing settings but not code?
- **New (Q12–Q22):**
  - Q12: NM rim lengths and node spacing.
  - Q13: minimum distance between teams' boxes in MRM.
  - Q14: what the "attacks" are.
  - Q15: drop test details.
  - Q16: whether the scored range must start at line 1.
  - Q17: line terminator, and whose USB stick is used.
  - Q18: whether a PC may feed Quick Mode.
  - Q19: what counts in the cost category.

Consultants

- **Anna Umbert** (spectrum regulation): which plots (power, hop pattern, clear-channel threshold) should go in the ANNEX for the 20 dBm hopping design?
- **Concepción Santos / Carles Puente** (antennas): placing a flex omni inside a plastic box on a 70 cm support; measuring its in-box gain at the lab; effect of the battery, the foil and hands.
- **Paradells / Perelló** (protocols, with the protocols team): one channel per team in MRM and listen-before-talk in NM.

- Q20: listen-before-talk for every team in MRM.
- Q21: the NM output file name for relay boxes.
- Q22 (18 Oct): whether the Pi 3's WiFi/BT chip may stay on the board disabled in firmware, or must be removed.

Full text in `radio/ATENEA_QUESTIONS.md`.

Files: `radio/MTP-F26_TeamC_Radio_LinkBudget.xlsx` (hand-in link budget, live formulas) · `radio/linkbudget.py` (same model, plus the MRM near-far printout) · `radio/phy_config.py` (radio profiles and checks) · `radio/fw/` (nRF24 driver and tests) · `radio/DECISIONS.md`, `HANDOFF_ET.md`, `NM_SPEC.md` (NM Interop Spec), `TEST_PLAN.md`, `ATENEA_QUESTIONS.md`. Repo: github.com/SantiBarrios2002/mtp-f26-teamc.

Primary sources: competition rules PDF (MTP F26, Network Upgrade III + Sustainability); Session #1 slides; AN5142 with the course annotations; CNAF 2017/2021 (BOE); nRF24L01+ Product Specification v1.0; Ebyte E01-ML01SP4, E01-2G4M27D and E01C manuals; Molex 146153 and Taoglas FXP73 antenna listings.